

Brain Stem

Department Anatomy
Neuroanatomy (3)

First Part

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Lectuer Points:

- Part of the brain stem.
- The brainstem has three broad functions.
- The brainstem anterior and posterior view.
- Cranial nerves and Brain stem.
- The medulla oblongata:
 - External features:
 - ✓ The anterior view of medulla oblongata.
 - ✓ The posterior surface of the medulla oblongata.
 - Internal features:
 - ✓ Transverse sections of the medulla oblongata at Level of decussation of the pyramids.
 - ✓ Transverse sections at Level of decussation of the medial lemnisci
 - ✓ Transverse sections at the level of the middle of the olivary nuclei.

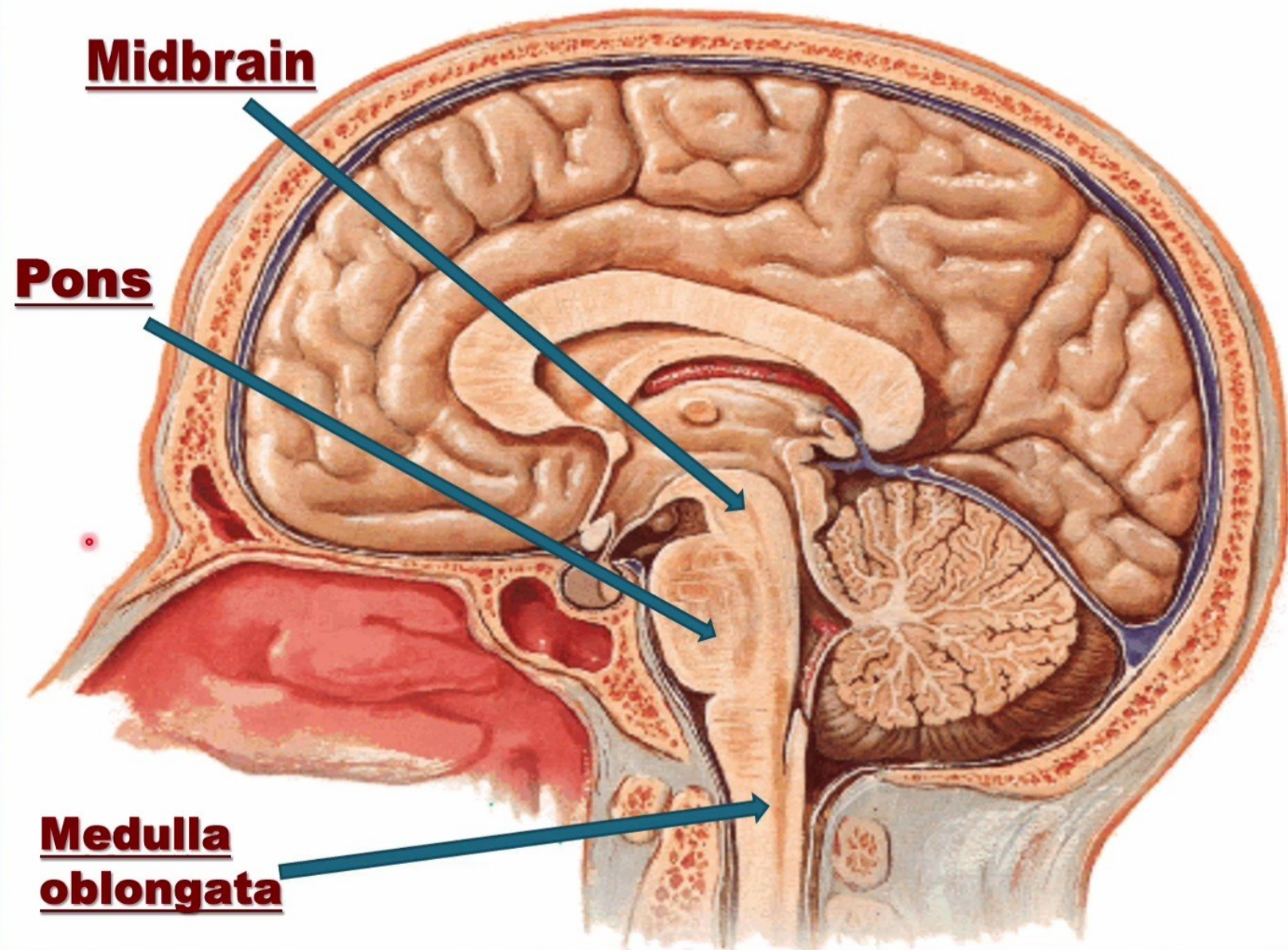
- The pons:
 - External features:
 - ✓ Anterior surface of the pons.
 - ✓ Posterior surface of the pons.
 - Internal features:
 - ✓ Transverse section through the caudal part of the pons at the level of the facial colliculus.
 - ✓ Transverse section through the pons at the level of the trigeminal nuclei
- The midbrain :
 - External features:
 - ✓ Anterior surface of the midbrain.
 - ✓ Posterior surface of the midbrain.
 - ✓ Lateral view of the midbrain.
 - Internal features:
 - ✓ Transverse section at the level of the superior colliculus.
 - ✓ Transverse sections of the midbrain at the level of the inferior colliculus.
- Summary of Anterior view of the Brain Stem.
- Summary of Posterior view of the Brain Stem.
- Summary of Cranial nerve nuclei in brain stem.
- Principal arteries of the brain stem (ventral view).
- Corticospinal (pyramidal) tract.

The brain stem

It is made up of:

- The midbrain.
- The pons.
- The medulla oblongata.

➤ It occupies the **posterior cranial fossa** of the skull.



The brainstem has three broad functions:

- 1) It serves as a conduit for the **ascending tracts** and **descending tracts (connecting the spinal cord to the different parts of the higher centers in the forebrain)**.
- 2) It contains important **reflex centers** associated with the control of **respiration** and the **cardiovascular system** and with the **control of consciousness**.
- 3) It contains the important **nuclei** of **cranial nerves III through XII**.

Cranial nerves and Brain stem

I **Olfactory**

II **Optic**

III **Oculomotor**

IV **Trochlear**

V **Trigeminal**

VI **Abducens**

VII **Facial**

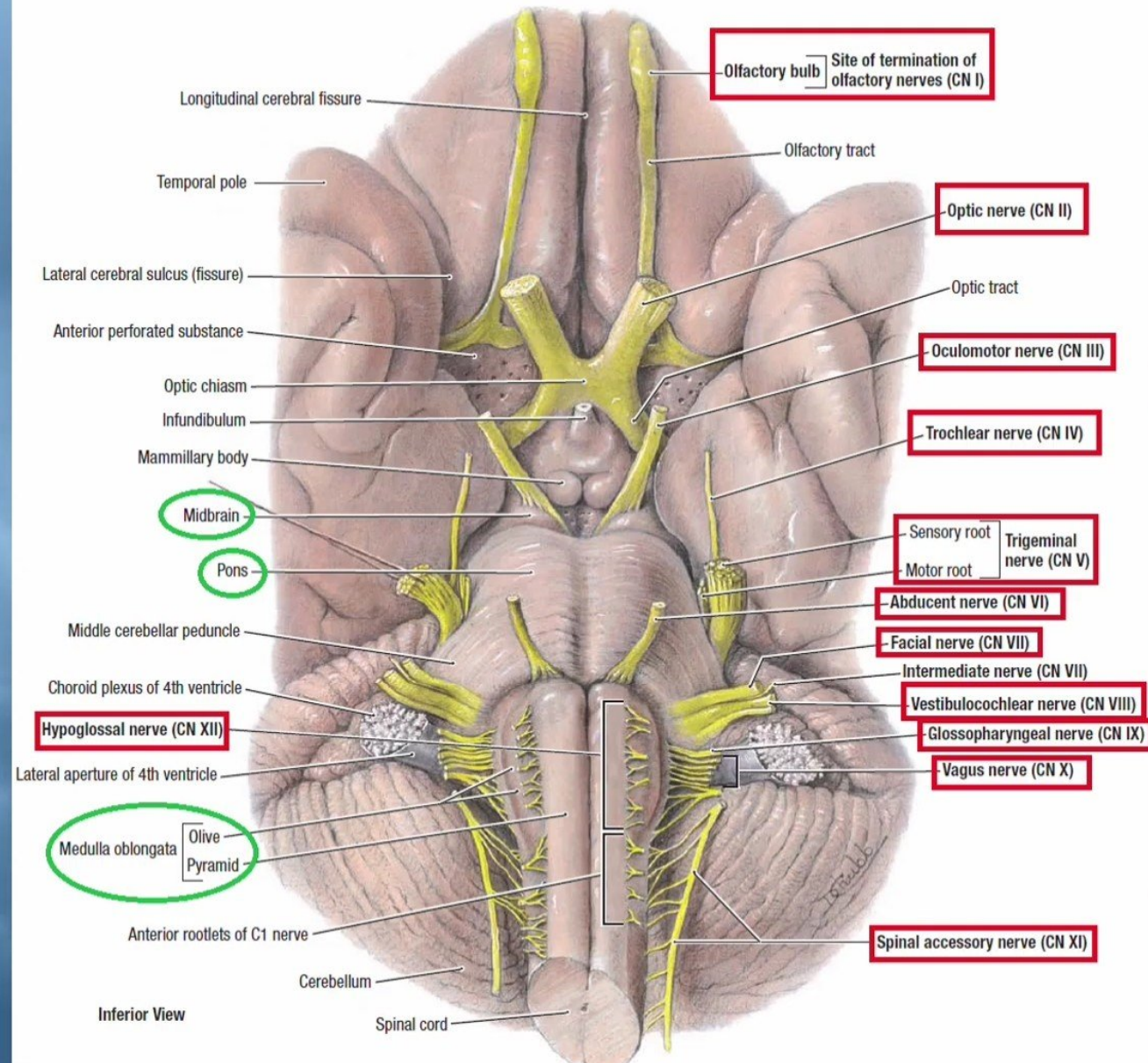
VIII **VestibuloCochlear**

IX **Glossopharyngeal**

X **Vagus**

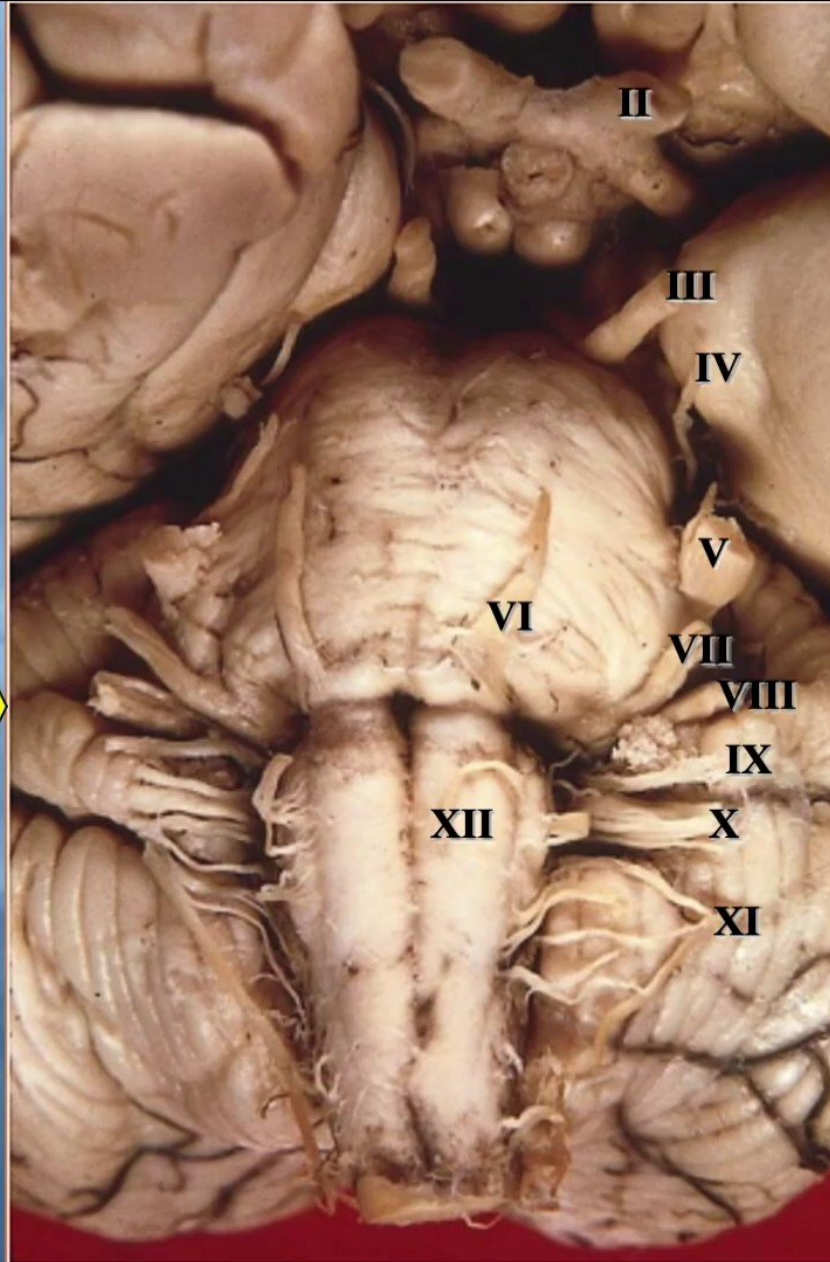
XI **Spinal accessory**

XII **Hypoglossal**



Cranial nerves and Brain stem

**CRANIAL
NERVES**



II **Optic**
(CNS Tract)

III **Oculomotor**

IV **Trochlear**

V **Trigeminal**

VI **Abducens**

VII **Facial**

VIII **VestibuloCochlear**

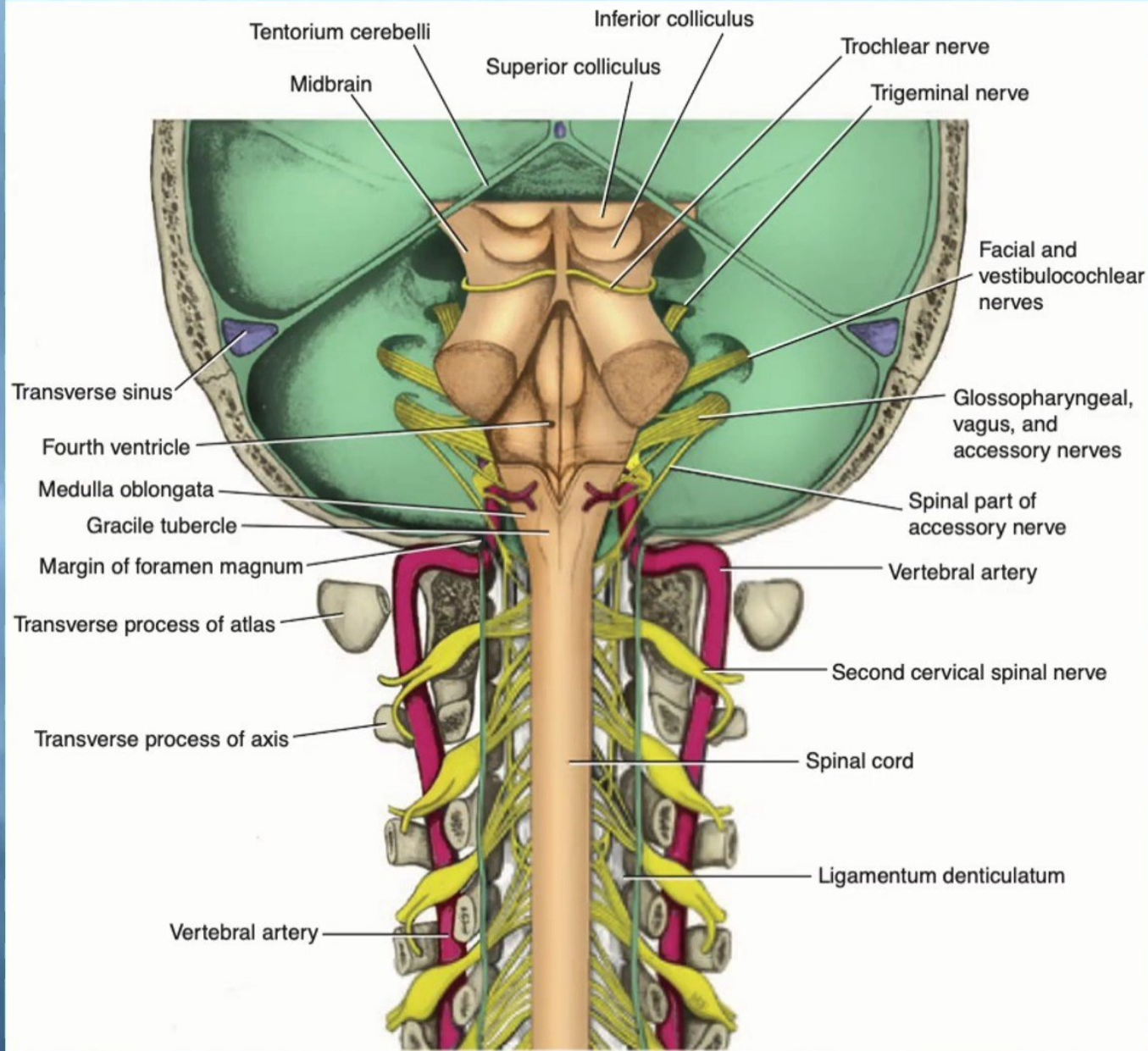
IX **Glossopharyngeal**

X **Vagus**

XI **Spinal accessory**

XII **Hypoglossal**

Posterior view of the brainstem after removal of the occipital and parietal bones and the cerebrum, the cerebellum, and the roof of the fourth ventricle. Laminae of the upper cervical vertebrae have also been removed.



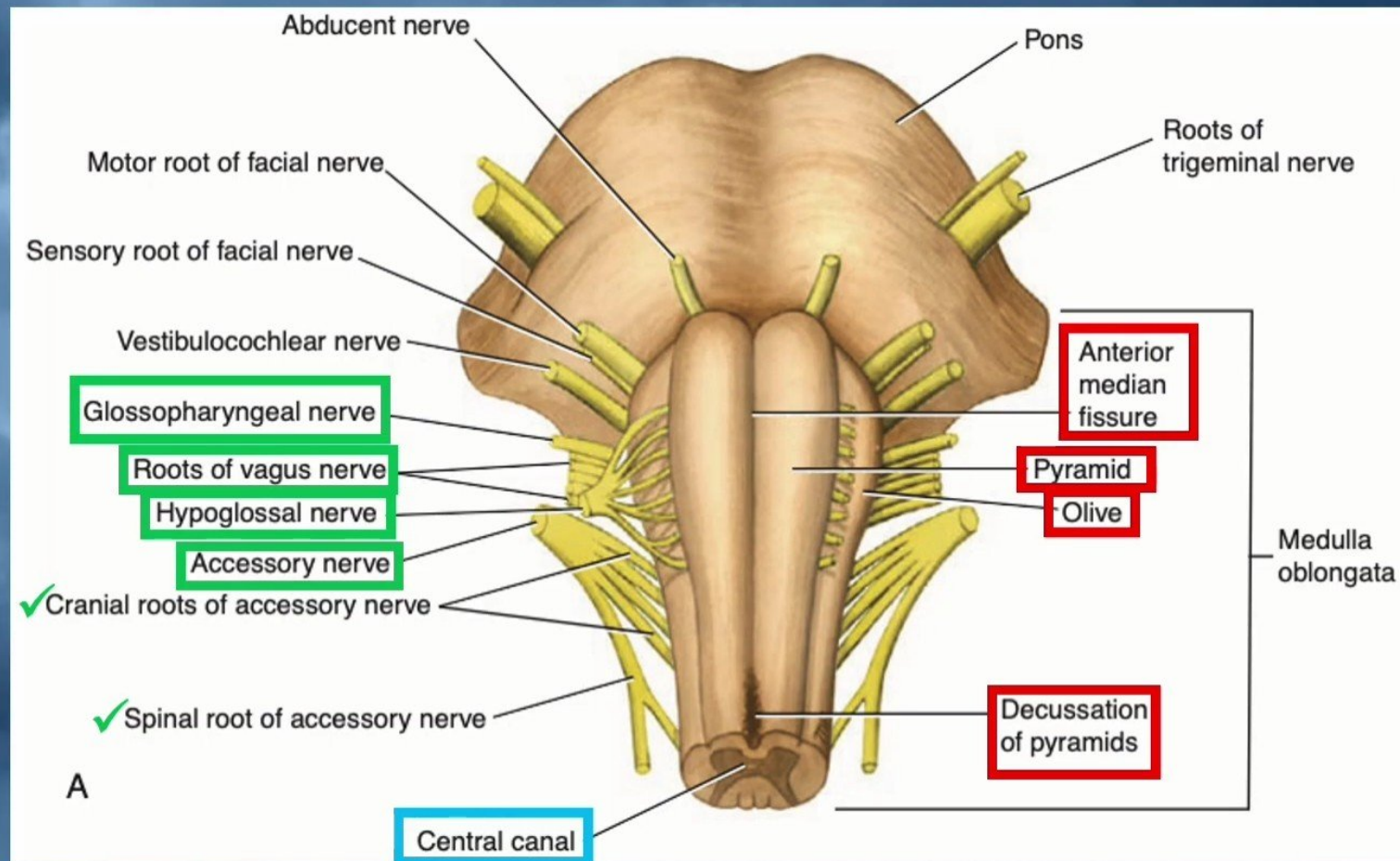
The anterior median fissure of the medulla :

- It is found on the anterior surface of the medulla which is continuous inferiorly with the **anterior median fissure** of the spinal cord.
- On each side of the median fissure, there is a swelling called the **pyramid**.

The pyramids:

- They are composed of **bundles of nerve fibers**, called **corticospinal fibers**.
- They **originate** in large nerve cells in the **precentral gyrus** of the cerebral cortex.
- Inferiorly, the descending fibers **cross** over to the opposite side, forming the **decussation of the pyramids**.

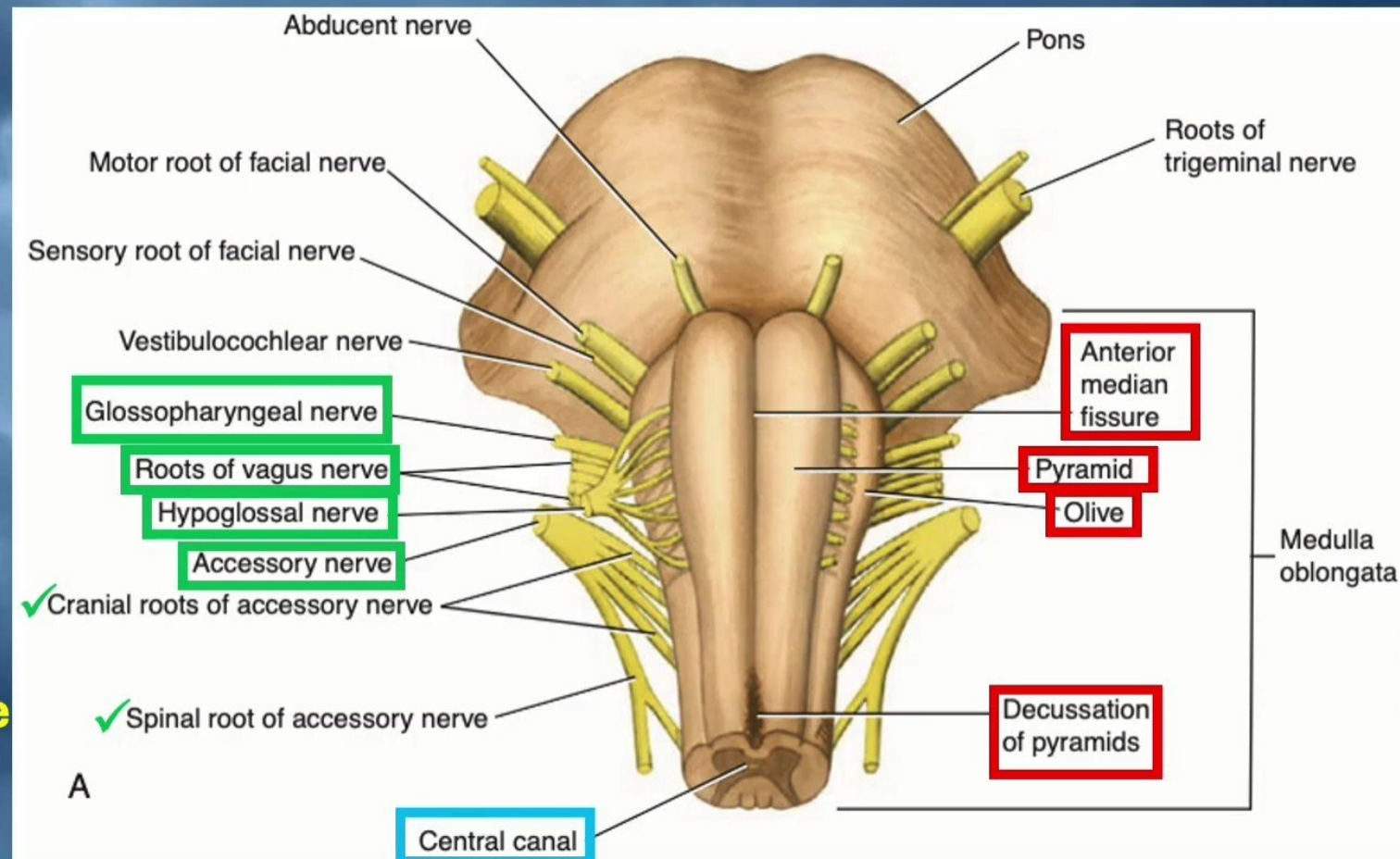
External features of the anterior view of medulla oblongata



The inferior cerebellar peduncles:

- They are found posterior to the **olives**.
- They connect the **medulla** to the **cerebellum**.
- In the groove between the **olive** and the **inferior cerebellar peduncle** emerge :
 - The **roots of the glossopharyngeal nerve**.
 - The **vagus nerve**.
 - The **cranial roots of the accessory nerve**.

External features Anterior view of medulla oblongata



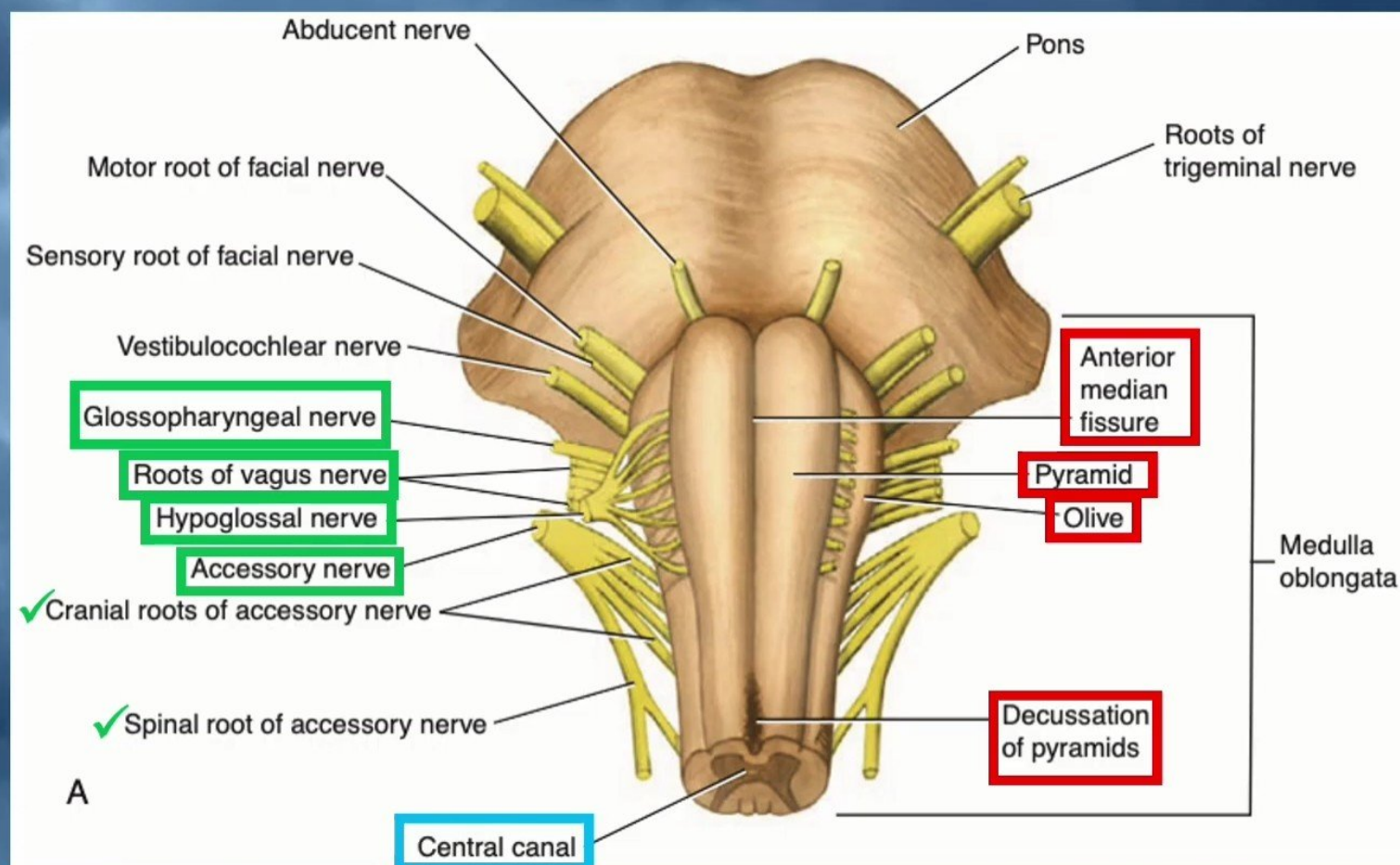
The anterior external arcuate fibers:

- They are a **few nerve fibers** that emerge from the **anterior median fissure** above the **decussation** and pass laterally over the surface of the medulla oblongata to enter the **cerebellum**.

The olives:

- It is found Posterolateral to the pyramids.
- They are **oval elevations**.
- They are produced by the underlying **inferior olivary nuclei**.
- In the groove between the **pyramid** and the **olive** emerge the **rootlets of the hypoglossal nerve**.

Anterior view of medulla oblongata



- They are formed the lower part of the **floor of the fourth ventricle**.

The posterior median sulcus:

- It is continuous with the posterior aspect of the spinal cord as **posterior median sulcus**.

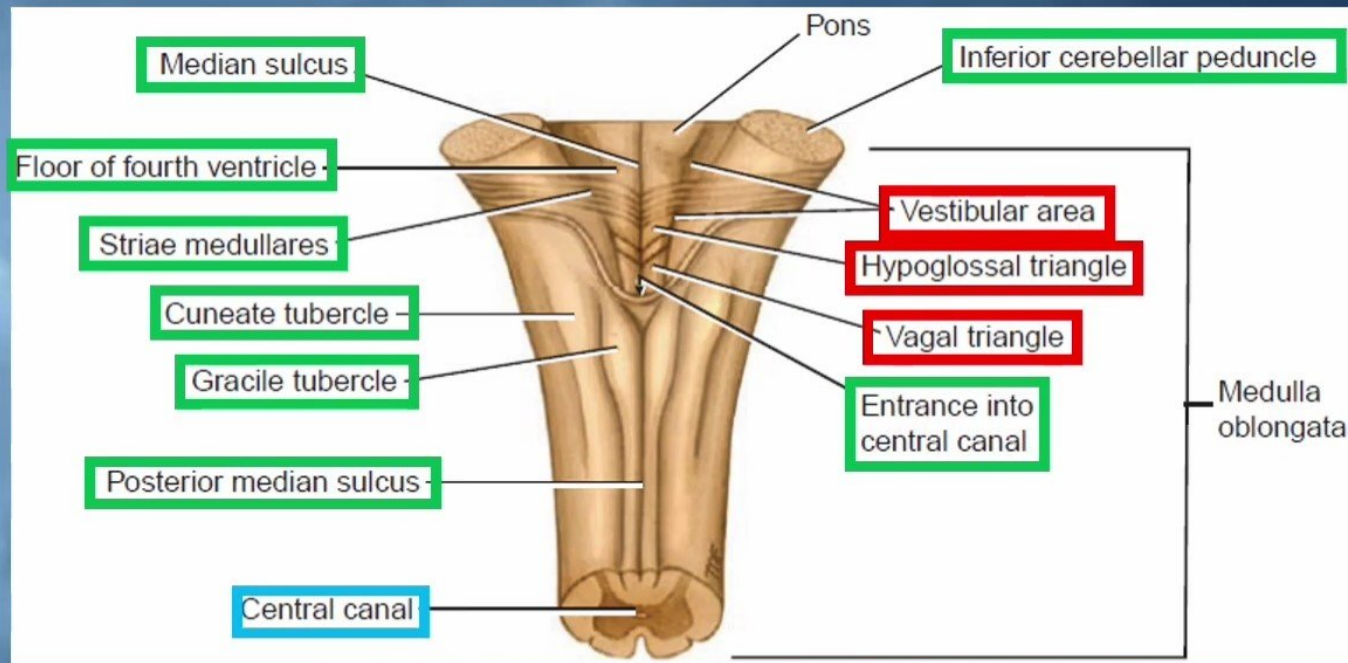
The gracile tubercle:

- On each side of the median sulcus, there is an elongated swelling, produced by the underlying **gracile nucleus**.

The cuneate tubercle:

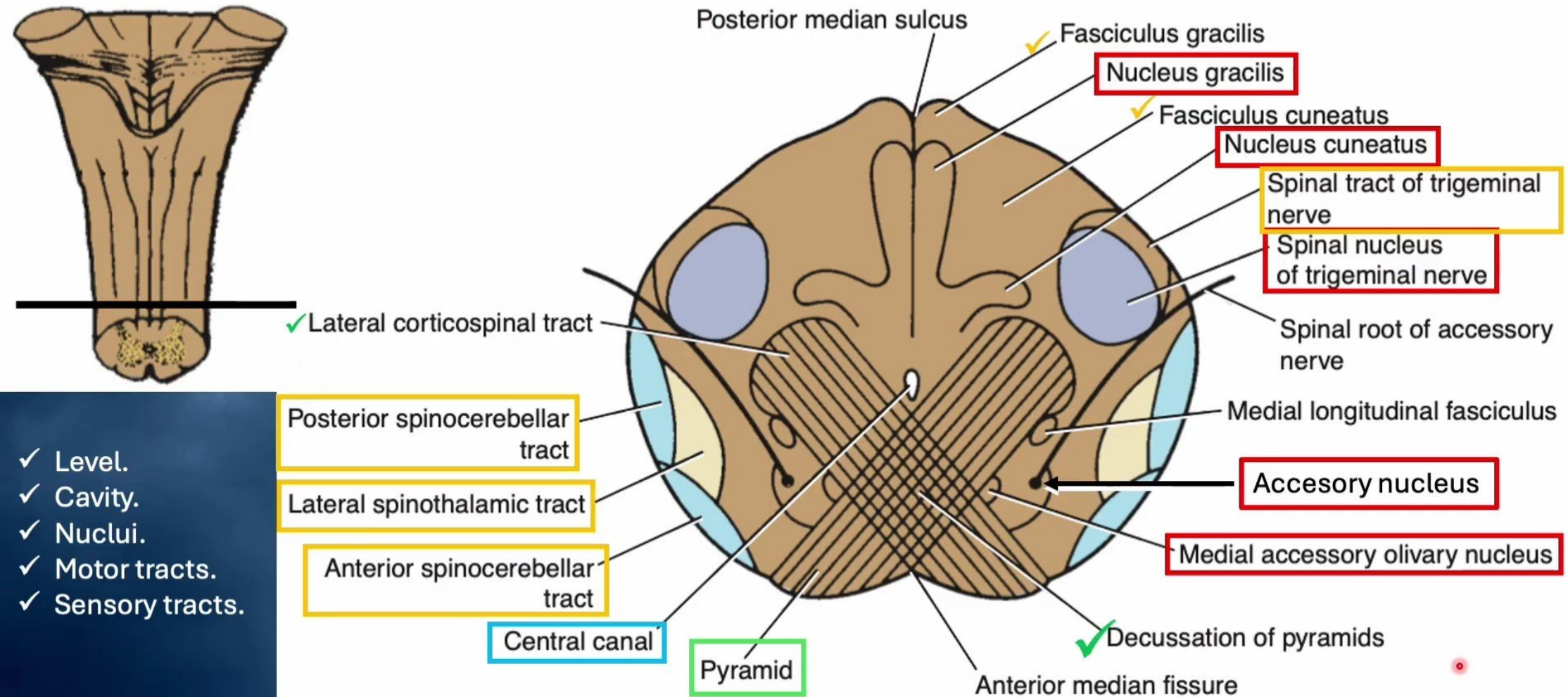
- Lateral to the gracile tubercle is a similar swelling,, produced by the underlying **cuneate nucleus**.

External features of the posterior surface of the medulla oblongata

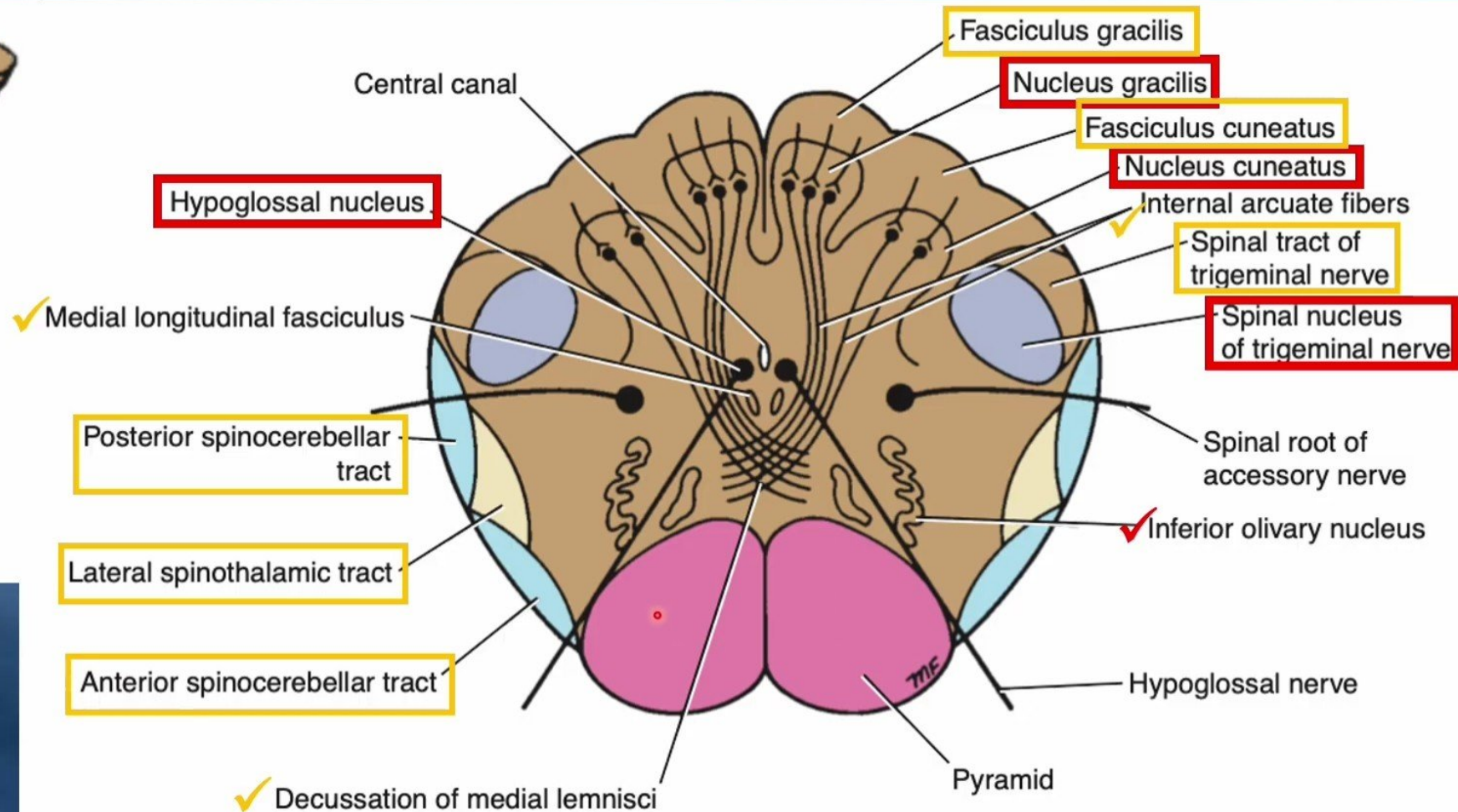
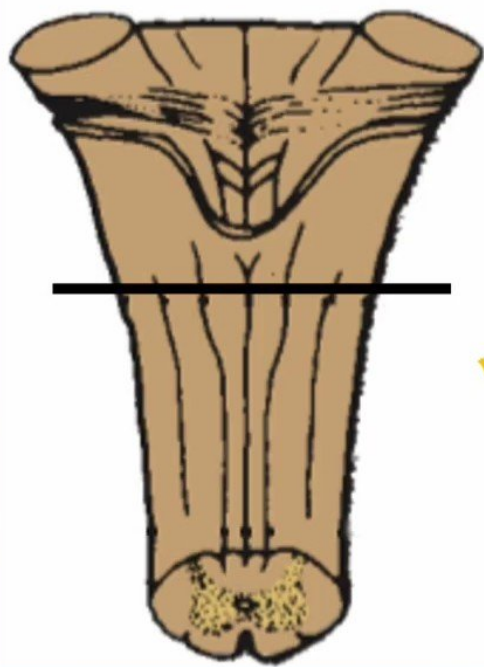


- **Note that the roof of the fourth ventricle and the cerebellum have been removed**

Transverse sections of the medulla oblongata at Level of decussation of the pyramids

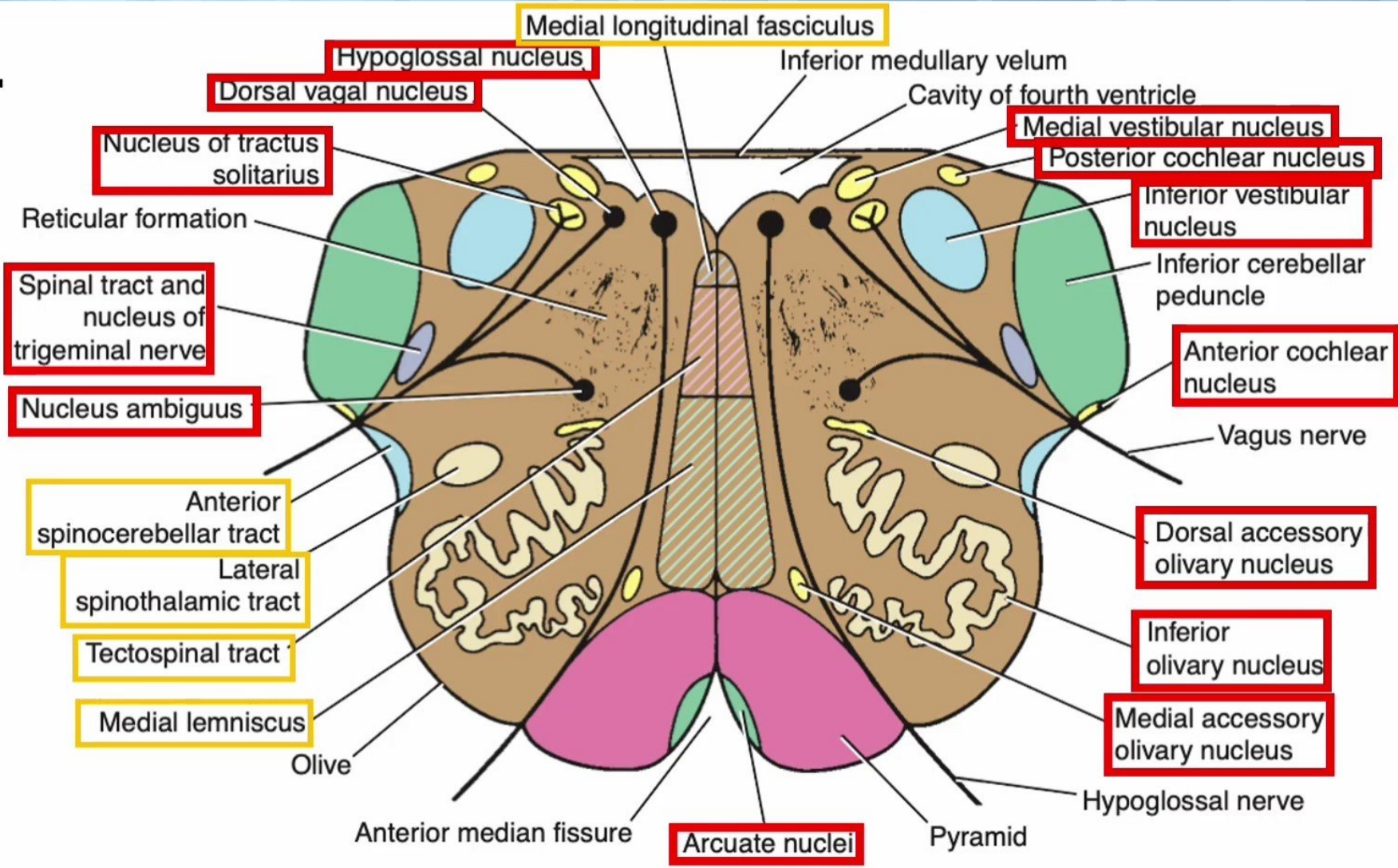
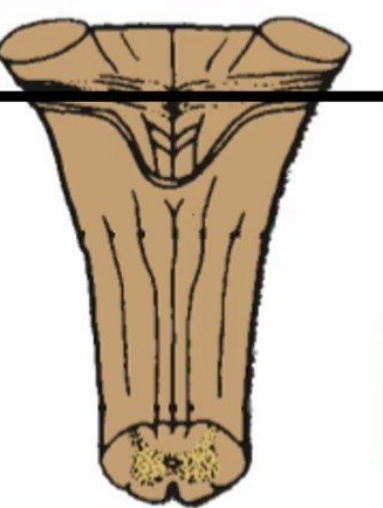


Transverse sections of the medulla oblongata at Level of decussation of the medial lemnisci



- ✓ Level.
- ✓ Cavity.
- ✓ Nuclui.
- ✓ Motor tracts.
- ✓ Sensory tracts.

Transverse sections of the medulla oblongata at the level of the middle of the olivary nuclei

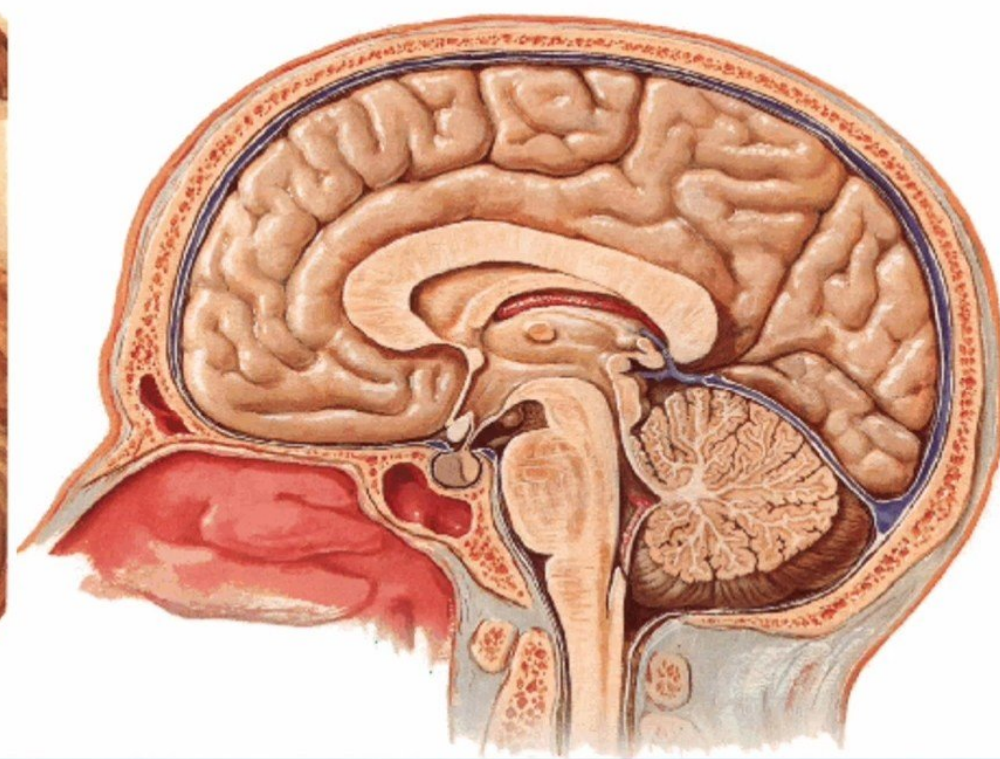
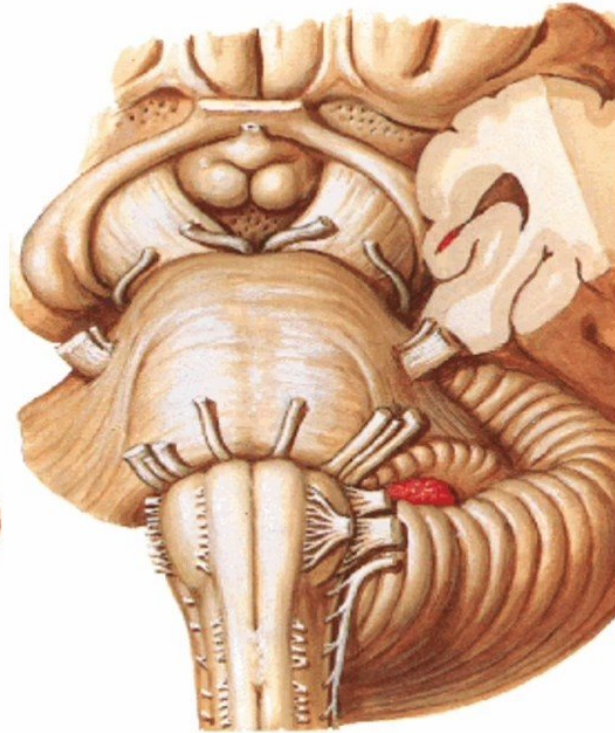
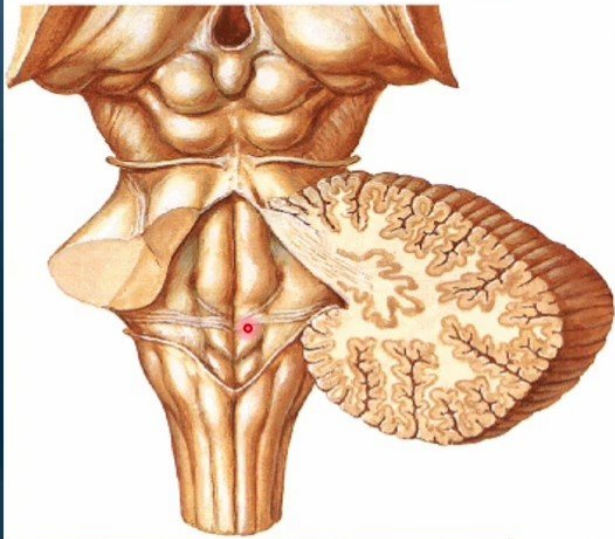


- ✓ Level.
- ✓ Cavity.
- ✓ Nuclui.
- ✓ Motor tracts.
- ✓ Sensory tracts.

Comparison of the Different Levels of the Medulla Oblongata Showing the Major Structures at Each Level^a

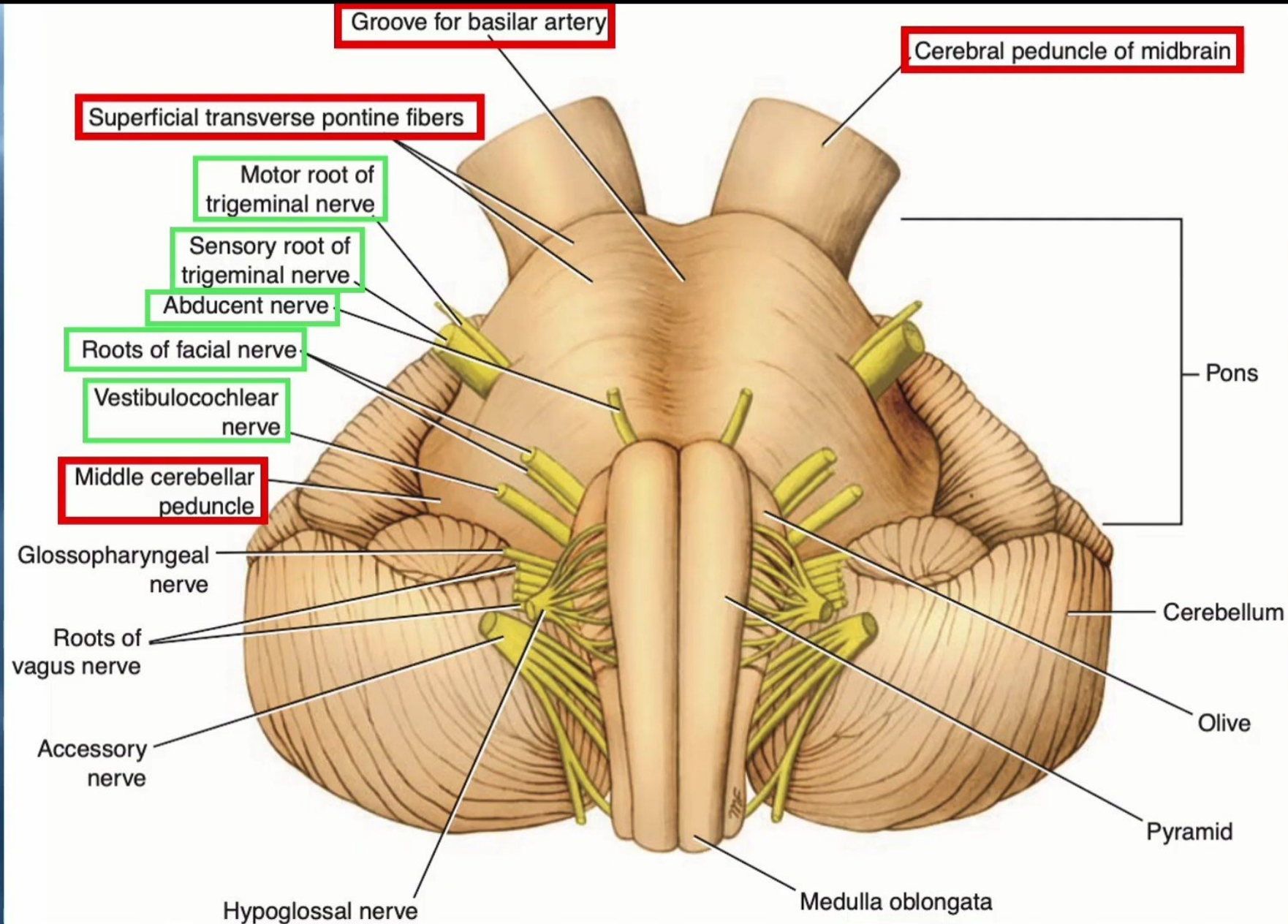
Level	Cavity	Nuclei	Motor Tracts	Sensory Tracts
Decussation of pyramids	Central canal	Nucleus gracilis, nucleus cuneatus, spinal nucleus of cranial nerve V, accessory nucleus	Decussation of corticospinal tracts, pyramids	Spinal tract of cranial nerve V, posterior spinocerebellar tract, lateral spinothalamic tract, anterior spinocerebellar tract
Decussation of medial lemnisci	Central canal	Nucleus gracilis, nucleus cuneatus, spinal nucleus of cranial nerve V, accessory nucleus, hypoglossal nucleus	Pyramids	Decussation of medial lemnisci, fasciculus gracilis, fasciculus cuneatus, spinal tract of cranial nerve V, posterior spinocerebellar tract, lateral spinothalamic tract, anterior spinocerebellar tract
Olives, inferior cerebellar peduncle	Fourth ventricle	Inferior olivary nucleus, spinal nucleus of cranial nerve V, vestibular nucleus, glossopharyngeal nucleus, vagal nucleus, hypoglossal nucleus, nucleus ambiguus, nucleus of tractus solitarius	Pyramids	Medial longitudinal fasciculus, tectospinal tract, medial lemniscus, spinal tract of cranial nerve V, lateral spinothalamic tract, anterior spinocerebellar tract

The pons

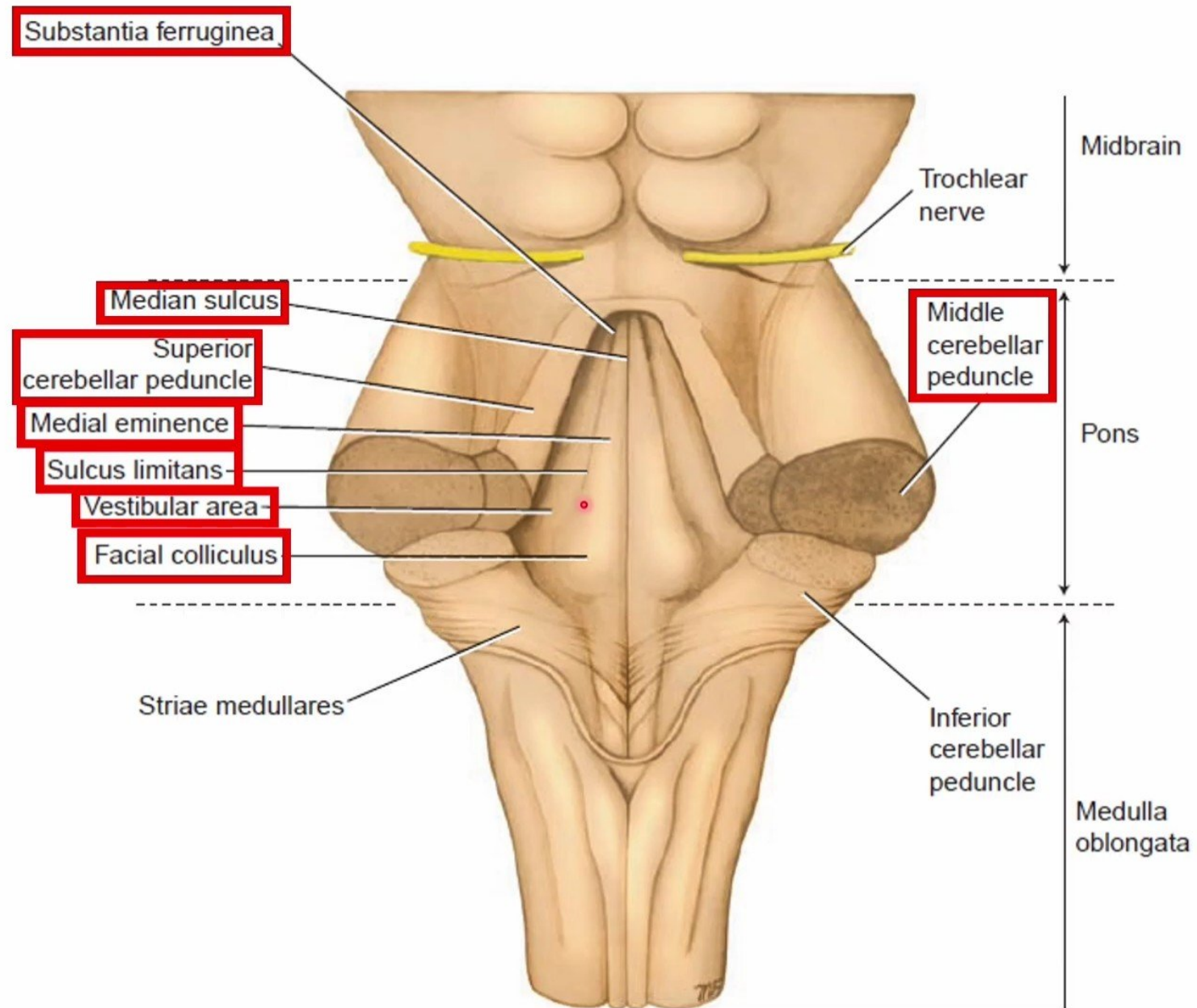


- It is **anterior** to the **cerebellum**.
- It connects the **medulla oblongata** to the **midbrain**.
- It is about **1 inch (2.5 cm) long**.
- The anterior surface is convex from side to side and shows many **transverse fibers** that converge on each side to form the **middle cerebellar peduncle**.
- There is a shallow groove in the midline, the **basilar groove**.

Anterior surface of the the pons



**Posterior surface
of the pons.
✓ The cerebellum
has been
removed.**



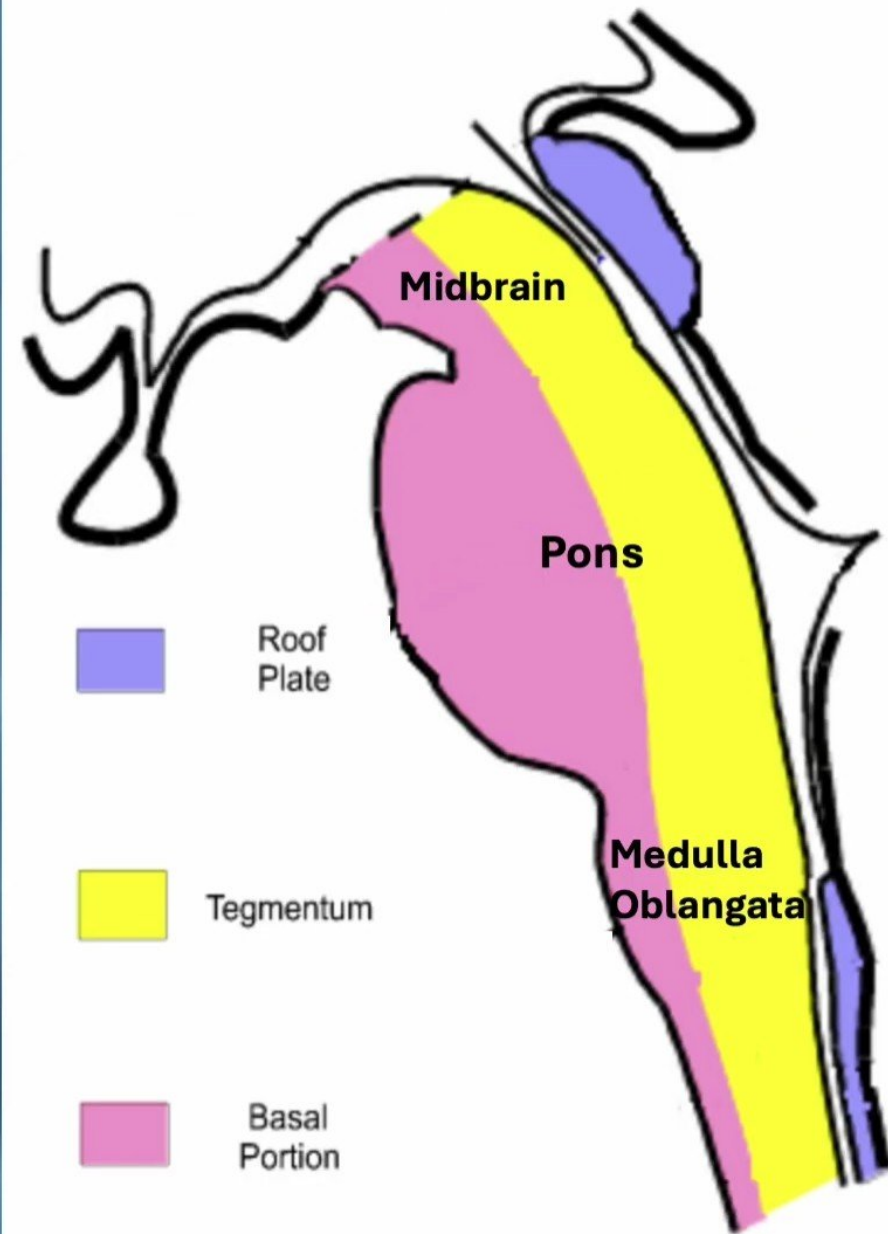
The pons

The pons is commonly divided into:

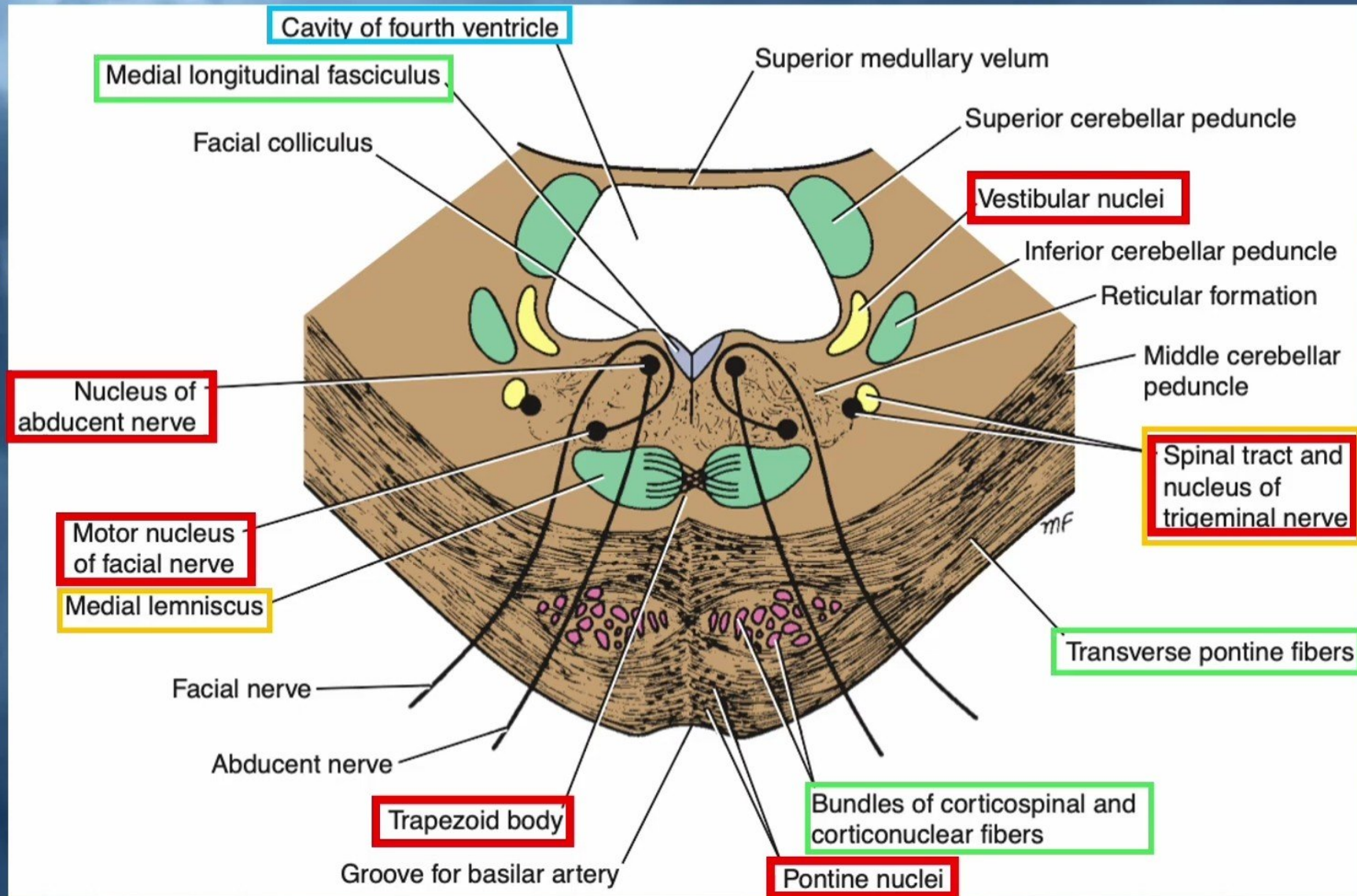
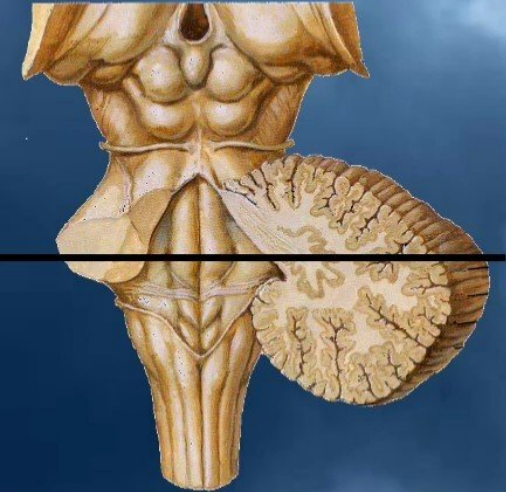
- ✓ Posterior part, the **tegmentum**.
- ✓ Anterior **basal part**. (form by the transverse pontine fiber).

The structure of the pons may be studied at two levels:

- (1) Transverse section through the caudal part, passing through **the facial colliculus**.
- (2) Transverse section through the cranial part, passing through **the trigeminal nuclei**.

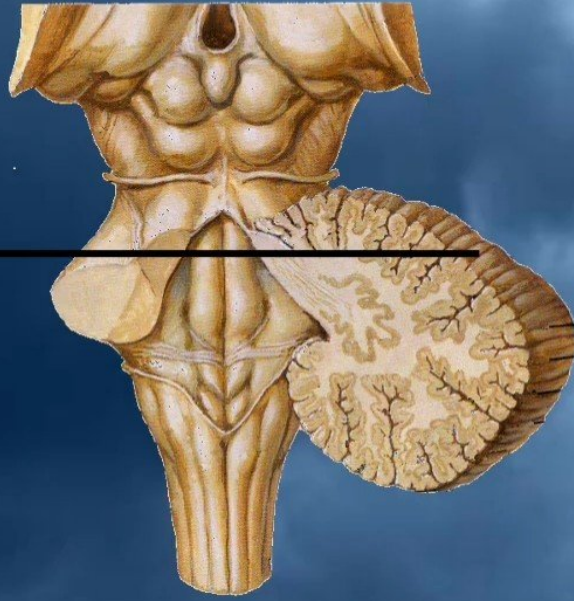


Transverse section through the caudal part of the pons at the level of the facial colliculus

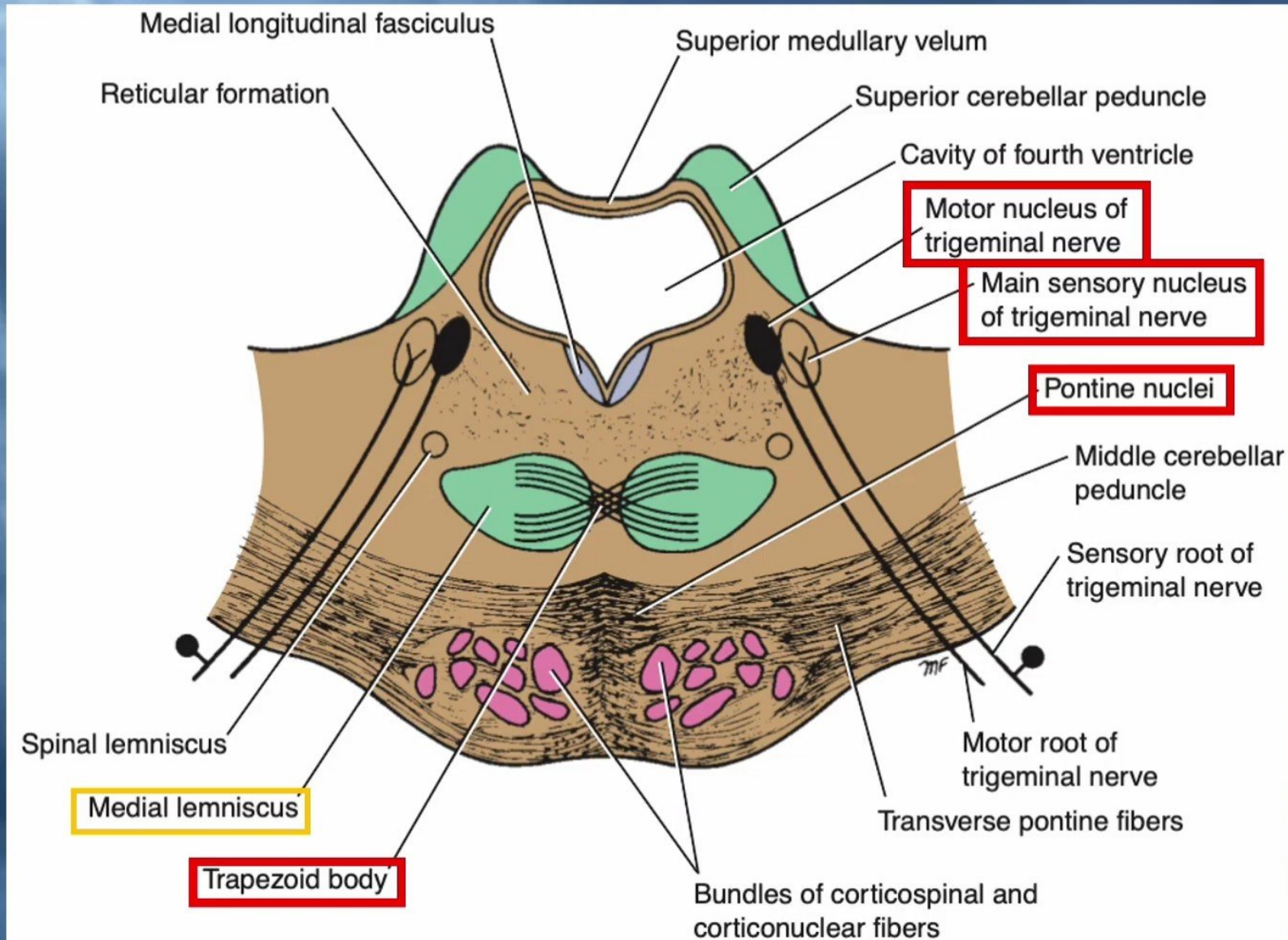


- ✓ Level.
- ✓ Cavity.
- ✓ Nuclui.
- ✓ Motor tracts.
- ✓ Sensory tracts.

Transverse section through the pons at the level of the trigeminal nuclei



- ✓ Cavity.
- ✓ Nuclui.
- ✓ Motor tracts.
- ✓ Sensory tracts.



The pons

Comparison of the Different Levels of the Pons Showing the Major Structures at Each Level^a

Level	Cavity	Nuclei	Motor Tracts	Sensory Tracts
Facial colliculus	Fourth ventricle	Facial nucleus, abducent nucleus, medial vestibular nucleus, spinal nucleus of cranial nerve V, pontine nuclei, trapezoid nuclei	Corticospinal and corticonuclear tracts, transverse pontine fibers, medial longitudinal fasciculus	Spinal tract of cranial nerve V; lateral, spinal, and medial lemnisci
Trigeminal nuclei	Fourth ventricle	Main sensory and motor nucleus of cranial nerve V, pontine nuclei, trapezoid nuclei	Corticospinal and corticonuclear tracts, transverse pontine fibers, medial longitudinal fasciculus	Lateral, spinal, and medial lemnisci

Will be continue...